```
      name:  SimulationBaltic
       log:  C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
> tput.smcl
  log type:  smcl
 opened on:  25 Aug 2026, 10:49:58

1 .
2 . asdoc, text(\par \qc Financial Sustainability of the Care for the Disabled: A Simula
  > tion of Benefits for Older People in the Baltic States up to 2050 with data from SHA
  > RE and Eurostat) fs(12)  save(report.doc) replace
  (file report.doc not found)
  Click to Open File:  report.doc

3 . asdoc, text(\par \qc Hans Gevers - Junior Research Fellow at the Estonian Business S
  > chool https://orcid.org/0009-0001-0249-4142 hans.gevers@ebs.ee) fs(10)  save(report.
  > doc) append
  Click to Open File:  report.doc

4 .
5 . *>>>PREPROCESSING
6 .
7 . foreach num of numlist 8(1)9{
    2.          clear all
    3.          cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
  > 7\sharew`num'_rel9-0-0_ALL_datasets_stata\"
    4.          use "sharew`num'_rel9-0-0_gv_imputations.dta"
    5.          keep if implicat==1
    6.          merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_health.dta"
    7.          isid mergeid
    8.          drop _merge
    9.          merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_weights.dta"
   10.          isid mergeid
   11.          drop _merge
   12.
8 .         if `num'==8{
   13.                  gen Qyear=2020
   14.          }
   15.          if `num'==9{
   16.                  gen Qyear=2022
   17.          }
   18.
9 .         keep if country==57 | country==48 | country==35
   19.
10.         save "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
  > \W`num'_final.dta", replace
   20. }
  C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew8_rel9-0-
  > 0_ALL_datasets_stata
  (214,780 observations deleted)
  (label orienti already defined)
  (label eurod already defined)
  (label numeracy already defined)
  (label gali already defined)
  (label language already defined)
  (label country already defined)

      Result                      Number of obs
      -----------------------------------------
      Not matched                             0
      Matched                            53,695  (_merge==3)
      -----------------------------------------

  (label country already defined)
  (label language already defined)
```

```
    Result                           Number of obs
    -----------------------------------------
    Not matched                                 0
    Matched                                53,695  (_merge==3)
    -----------------------------------------

(47,632 observations deleted)
file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
    7\W8_final.dta saved
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew9_rel9-0-
> 0_ALL_datasets_stata
(277,788 observations deleted)
(label orienti already defined)
(label eurod already defined)
(label numeracy already defined)
(label gali already defined)
(label language already defined)
(label country already defined)

    Result                           Number of obs
    -----------------------------------------
    Not matched                                 0
    Matched                                69,447  (_merge==3)
    -----------------------------------------

(label country already defined)
(label language already defined)

    Result                           Number of obs
    -----------------------------------------
    Not matched                                 0
    Matched                                69,447  (_merge==3)
    -----------------------------------------

(61,831 observations deleted)
file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
    7\W9_final.dta saved

11.
12. clear all

13. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7

14. use "W8_final.dta"

15. merge 1:1 mergeid using "weightslong.dta", keep(match)
(label country already defined)

    Result                           Number of obs
    -----------------------------------------
    Not matched                                 0
    Matched                                 5,101  (_merge==3)
    -----------------------------------------

16. drop _merge

17. save "W8_final.dta", replace
file W8_final.dta saved

18.
19. clear all
```

```
20. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
  C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7

21. use "W9_final.dta"

22. merge 1:1 mergeid using "weightslong.dta", keep(match)
   (label country already defined)

    Result                           Number of obs
    ─────────────────────────────────────────────
    Not matched                                  0
    Matched                                  4,921  (_merge==3)
    ─────────────────────────────────────────────

23. drop _merge

24. save "W9_final.dta", replace
  file W9_final.dta saved

25.
26. clear all

27. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
  C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7

28. use "W8_final.dta"

29. append using "W9_final.dta"
   (variable psu was str10, now str11 to accommodate using data's values)
   (variable ssu was str6, now str13 to accommodate using data's values)
   (label pstat already defined)
   (label language already defined)
   (label country already defined)
   (label Hrespt already defined)
   (label pcskill already defined)
   (label IMPflag already defined)
   (label orienti already defined)
   (label bfi10 already defined)
   (label thospital already defined)
   (label satisfied already defined)
   (label often already defined)
   (label yedu already defined)
   (label fdistress already defined)
   (label writing already defined)
   (label reading already defined)
   (label hearing already defined)
   (label eyesight already defined)
   (label mstat already defined)
   (label sphus already defined)
   (label noyes already defined)
   (label hnrsc already defined)
   (label undersq already defined)
   (label clarif already defined)
   (label willans already defined)
   (label otrf already defined)
   (label empstat already defined)
   (label cjs already defined)
   (label isced already defined)
   (label gender already defined)
   (label num already defined)
   (label gali already defined)
   (label meat already defined)
   (label legeggs already defined)
   (label fruit already defined)
   (label numeracy already defined)
   (label numeracy2 already defined)
   (label eurod already defined)
   (label dairyp already defined)
   (label weekday already defined)
   (label specday already defined)
   (label dkrf already defined)
   (label dummi already defined)
```

```
  (label zero_one already defined)
  (label loneli already defined)
  (label chronic2 already defined)
  (label sphus2 already defined)
  (label bmi2 already defined)
  (label mobilit2 already defined)
  (label mobilit3 already defined)
  (label adl2 already defined)
  (label iadl2 already defined)
  (label phactiv already defined)

30.
31. save "dataset.dta", replace
  file dataset.dta saved

32.
33. clear all

34.
35. use "dataset.dta"

36.
37. *>>>PREPARATION
38.
39. label define isc 0 "None" 1 "Primary education" 2 "Lower secondary education" 3 "Upp
  > er secondary education" 4 "Post-secondary non-tertiary education" 5 "Bachelor's or e
  > quivalent level" 6 "Master's or equivalent level/Doctoral or equivalent level"

40. label values isced isc

41. label variable isced "Education"

42.
43. generate income = thinc-ypen3

44.
45. egen id= group(mergeid)

46.
47. drop if cjs==-99
  (70 observations deleted)

48.
49. generate educ=0

50. replace educ=1 if isced==3 | isced==4
  (5,234 real changes made)

51. replace educ=2 if isced>=5
  (2,943 real changes made)

52. label define educL 0 "Less than primary, primary and lower secondary education (ISCE
  > D levels 0-2)" 1 "Upper secondary and post-secondary non-tertiary education (ISCED l
  > evels 3 and 4)" 2 "Tertiary education (ISCED levels 5-8)"

53. label values educ educL

54.
55. keep single ypen3 gender cjs age educ sphus country income gali id Qyear sphus
```

```
56.
57. *>>>DESCRIPTIVES
58.
59. tabulate country

    Country |
 identifier |      Freq.     Percent        Cum.
------------+-----------------------------------
    Estonia |      5,526       55.53       55.53
  Lithuania |      2,423       24.35       79.87
     Latvia |      2,003       20.13      100.00
------------+-----------------------------------
      Total |      9,952      100.00

60. hist ypen3
  (bin=39, start=0, width=333.33333)

61.
62. asdoc codebook single ypen3 gender cjs age educ sphus country income gali id Qyear,
  > compact save(report.doc) append

 Variable     Obs Unique      Mean        Min      Max  Label
----------------------------------------------------------------------------------------
  --------------------------------------------------------------------------------------
  ------------------------------
 single      9952      2  .4113746          0        1  Single
 ypen3       9952    383  238.8071          0    13000  Disability pensions/benefits
 gender      9952      2  1.631732          1        2  Gender
 cjs         9952      6   1.94996          1       97  Current job situation
 age         9952     64  69.68238         34      100  Age
 educ        9952      3  1.117363          0        2
 sphus       9952      5  3.740856          1        5  Self-perceived health - US scale
 country     9952      3  42.59295         35       57  Country identifier
 income      9952   4456  10092.82  -.0000737   170698
 gali        9952      2   .588324          0        1  Limitation with activities
 id          9952   5086  2559.018          1     5101  group(mergeid)
 Qyear       9952      2   2020.98       2020     2022
----------------------------------------------------------------------------------------
  --------------------------------------------------------------------------------------
  ------------------------------
 Click to Open File:  report.doc

63.
64. asdoc codebook, save(report.doc) append

----------------------------------------------------------------------------------------
 country                                                              Country identifier
----------------------------------------------------------------------------------------

                   Type: Numeric (byte)
                  Label: country

                  Range: [35,57]                         Units: 1
          Unique values: 3                           Missing .: 0/9,952

             Tabulation: Freq.   Numeric  Label
                         5,526        35  Estonia
                         2,423        48  Lithuania
                         2,003        57  Latvia

----------------------------------------------------------------------------------------
 single                                                                           Single
----------------------------------------------------------------------------------------

                   Type: Numeric (byte)
                  Label: noyes

                  Range: [0,1]                           Units: 1
          Unique values: 2                           Missing .: 0/9,952
```

```
           Tabulation: Freq.   Numeric  Label
                       5,858         0  No
                       4,094         1  Yes

-------------------------------------------------------------------------------
ypen3                                            Disability pensions/benefits
-------------------------------------------------------------------------------

                 Type: Numeric (double)

                Range: [0,13000]                    Units: 1.000e-07
        Unique values: 383                      Missing .: 0/9,952

                 Mean: 238.807
            Std. dev.: 914.444

          Percentiles:     10%       25%       50%       75%       90%
                             0         0         0         0       276

-------------------------------------------------------------------------------
gender                                                                   Gender
-------------------------------------------------------------------------------

                 Type: Numeric (byte)
                Label: gender

                Range: [1,2]                        Units: 1
        Unique values: 2                        Missing .: 0/9,952

           Tabulation: Freq.   Numeric  Label
                       3,665         1  Male
                       6,287         2  Female

-------------------------------------------------------------------------------
age                                                                         Age
-------------------------------------------------------------------------------

                 Type: Numeric (int)
                Label: num, but 64 nonmissing values are not labeled

                Range: [34,100]                     Units: 1
        Unique values: 64                       Missing .: 0/9,952

             Examples: 60
                       66
                       72
                       80

-------------------------------------------------------------------------------
sphus                                         Self-perceived health - US scale
-------------------------------------------------------------------------------

                 Type: Numeric (byte)
                Label: sphus

                Range: [1,5]                        Units: 1
        Unique values: 5                        Missing .: 0/9,952

           Tabulation: Freq.   Numeric  Label
                         154         1  Excellent
                         416         2  Very good
                       2,844         3  Good
                       4,979         4  Fair
                       1,559         5  Poor

-------------------------------------------------------------------------------
gali                                               Limitation with activities
-------------------------------------------------------------------------------

                 Type: Numeric (byte)
                Label: gali
```

```
          Range: [0,1]                      Units: 1
  Unique values: 2                      Missing .: 0/9,952

    Tabulation: Freq.   Numeric  Label
                4,097         0  Not limited
                5,855         1  Limited
```

---

cjs — **Current job situation**

```
           Type: Numeric (byte)
          Label: cjs

          Range: [1,97]                     Units: 1
  Unique values: 6                      Missing .: 0/9,952

    Tabulation: Freq.   Numeric  Label
                6,182         1  Retired
                2,948         2  Employed or self-employed
                  241         3  Unemployed
                  440         4  Permanently sick
                   96         5  Homemaker
                   45        97  Other
```

Qyear — **(unlabeled)**

```
           Type: Numeric (float)

          Range: [2020,2022]                Units: 1
  Unique values: 2                      Missing .: 0/9,952

    Tabulation: Freq.  Value
                5,076  2020
                4,876  2022
```

---

income — **(unlabeled)**

```
           Type: Numeric (float)

          Range: [-.00007374,170697.95]     Units: 1.000e-20
  Unique values: 4,456                  Missing .: 0/9,952

           Mean: 10092.8
      Std. dev.: 8969.56

    Percentiles:      10%       25%       50%       75%       90%
                  1999.55      4920      7800   12895.6     20000
```

---

id — **group(mergeid)**

```
           Type: Numeric (float)

          Range: [1,5101]                   Units: 1
  Unique values: 5,086                  Missing .: 0/9,952

           Mean: 2559.02
      Std. dev.: 1474.17

    Percentiles:      10%       25%       50%       75%       90%
                      513    1281.5    2563.5    3837.5      4596
```

---

educ — **(unlabeled)**

```
                Type: Numeric (float)
               Label: educL

               Range: [0,2]                        Units: 1
       Unique values: 3                        Missing .: 0/9,952

          Tabulation: Freq.   Numeric  Label
                      1,775         0  Less than primary, primary and
                                       lower secondary education
                                       (ISCED levels 0-2)
                      5,234         1  Upper secondary and
                                       post-secondary non-tertiary
                                       education (ISCED levels 3 and
                                       4)
                      2,943         2  Tertiary education (ISCED
                                       levels 5-8)
  Click to Open File:  report.doc

65.
66. vioplot age, over(gender) over(country) scheme(s2color) ytitle("Number of respondent
  > s", margin(small)) ylabel(30(10)100,angle(0) labsize(small) grid)

67. graph export sample.png, replace height(2400)
  (file sample.png not found)
  file sample.png saved as PNG format

68. vioplot age if ypen3>0, over(gender) over(country) scheme(s2color) ytitle("Number of
  >  respondents", margin(small)) ylabel(30(10)100,angle(0) labsize(small) grid)

69. graph export sampleDis.png, replace height(2400)
  (file sampleDis.png not found)
  file sampleDis.png saved as PNG format

70.
71. asdoc spearman single ypen3 gender cjs age educ sphus country income gali id Qyear s
  > phus, save(report.doc) append

  Number of observations = 9,952

               |   single    ypen3   gender      cjs      age     educ    sphus  country
  >    income  |  gali       id    Qyear    sphus
  -------------+------------------------------------------------------------------------
    ---------------------------------------------

        single |   1.0000
         ypen3 |  -0.0096   1.0000
        gender |   0.3013  -0.0183   1.0000
           cjs |  -0.1685   0.2522  -0.0665   1.0000
           age |   0.2603  -0.1646   0.0716  -0.7429   1.0000
          educ |  -0.1051  -0.0669   0.0455   0.1544  -0.2074   1.0000
         sphus |   0.0935   0.2086  -0.0050  -0.2445   0.3377  -0.2319   1.0000
       country |  -0.0445  -0.0944  -0.0115   0.0277  -0.0851   0.0267  -0.0232   1.0000
  >
        income |  -0.4997  -0.1257  -0.1489   0.0389  -0.0763   0.2033  -0.1560  -0.3441
  >    1.0000
          gali |   0.0865   0.2264   0.0208  -0.2098   0.2909  -0.1698   0.5394  -0.0199
  >   -0.1441   1.0000
            id |  -0.0448  -0.0924  -0.0068   0.0230  -0.0729   0.0289  -0.0243   0.8979
  >   -0.3002  -0.0243   1.0000
         Qyear |   0.0286  -0.0337   0.0136  -0.0629   0.0851   0.0045   0.0147   0.0042
  >    0.0789   0.0259   0.0038   1.0000
         sphus |   0.0935   0.2086  -0.0050  -0.2445   0.3377  -0.2319   1.0000  -0.0232
  >   -0.1560   0.5394  -0.0243   0.0147   1.0000
  Click to Open File:  report.doc
```

```
72. 
73. *>>>ANALYSIS
74. 
75. xtset id Qyear

  Panel variable: id (unbalanced)
   Time variable: Qyear, 2020 to 2022, but with gaps
           Delta: 1 unit

76. 
77. *-------
78. *ESTONIA
79. 
80. *general model
81. 
82. xttobit ypen3 i.gender age i.gali i.educ income if country==35, ll(0) iterate(25) ba
  > selevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -13059.972
  Iteration 1:  Log likelihood = -8969.4753
  Iteration 2:  Log likelihood = -8848.5585
  Iteration 3:  Log likelihood = -8845.8105
  Iteration 4:  Log likelihood = -8845.8086
  Iteration 5:  Log likelihood = -8845.8086

  Fitting full model:

  Iteration 0:  Log likelihood = -12748.166
  Iteration 1:  Log likelihood = -8956.6397
  Iteration 2:  Log likelihood = -8402.0038
  Iteration 3:  Log likelihood = -8347.2466
  Iteration 4:  Log likelihood = -8346.5547
  Iteration 5:  Log likelihood = -8346.5545

  Obtaining starting values for full model:

  Iteration 0:  Log likelihood = -45249.222
  Iteration 1:  Log likelihood = -45212.213
  Iteration 2:  Log likelihood = -45211.856
  Iteration 3:  Log likelihood = -45211.856

  Fitting full model:

  Iteration 0:  Log likelihood = -11974.249
  Iteration 1:  Log likelihood = -9486.0836  (not concave)
  Iteration 2:  Log likelihood =  -8892.275
  Iteration 3:  Log likelihood = -8211.8878
  Iteration 4:  Log likelihood = -8201.1002
  Iteration 5:  Log likelihood = -8198.0475
  Iteration 6:  Log likelihood = -8198.0437
  Iteration 7:  Log likelihood = -8198.0437

  Random-effects tobit regression                Number of obs     =  5,526
                                                        Uncensored =    767
  Limits: Lower =    0                               Left-censored =  4,759
          Upper = +inf                              Right-censored =      0

  Group variable: id                             Number of groups  =  2,837
  Random effects u_i ~ Gaussian                  Obs per group:
                                                               min =      1
                                                               avg =    1.9
                                                               max =      2

  Integration method: mvaghermite                Integration pts.  =     12

                                                 Wald chi2(6)      = 429.43
  Log likelihood = -8198.0437                    Prob > chi2       = 0.0000
```

```
─────────────────────────────────────────────────────────────────────────────────┬───
  ─────────────────────────────────────────────────────────────
                                                                           ypen3 │ Co
> efficient
>           Std. err.
>                           z
>                                P>|z|
>                                          [95% con
>                                                  f. interval]
─────────────────────────────────────────────────────────────────────────────────┼───
  ─────────────────────────────────────────────────────────────
                                                                          gender │
                                                                            Male │
>        0
>           (base)
                                                                          Female │
> 67.89776
>            173.9935
>                         0.39
>                                0.696
>                                        -273.1233
>                                                    408.9188
                                                                             age │  -
> 141.9375
>            8.822652
>                       -16.09
>                                0.000
>                                        -159.2296
>                                                   -124.6454
                                                                                 │
                                                                            gali │
                                                                     Not limited │
>        0
>           (base)
                                                                         Limited │
> 3129.336
>            212.4268
>                        14.73
>                                0.000
>                                         2712.987
>                                                    3545.685
                                                                                 │
                                                                            educ │
   Less than primary, primary and lower secondary education (ISCED levels 0-2) │
>        0
>           (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) │
> -653.254
>            217.7769
>                        -3.00
>                                0.003
>                                        -1080.089
>                                                    -226.419
                                         Tertiary education (ISCED levels 5-8) │  -
> 1353.043
>            265.9793
>                        -5.09
>                                0.000
>                                        -1874.353
>                                                   -831.7329
                                                                                 │
                                                                          income │  -
> .0763606
>            .0097108
>                        -7.86
>                                0.000
>                                        -.0953934
>                                                   -.0573278
                                                                           _cons │
> 5497.181
>            664.9515
```

```
>                        8.27
>                               0.000
>                                           4193.9
>                                                     6800.462
─────────────────────────────────────────────────────────────────────────┼───
  ──────────────────────────────────────────────────────
                                                              /sigma_u |
>  2512.79
>          113.4388
>                     22.15
>                               0.000
>                                         2290.454
>                                                     2735.126
                                                              /sigma_e |
> 1997.686
>          73.26852
>                     27.27
>                               0.000
>                                         1854.083
>                                                      2141.29
─────────────────────────────────────────────────────────────────────────┼───
  ──────────────────────────────────────────────────────
                                                                  rho |
> .6127313
>          .0271584
>                                         .5586039
>                                                     .6647491
─────────────────────────────────────────────────────────────────────────┴───
  ──────────────────────────────────────────────────────
LR test of sigma_u=0: chibar2(01) = 297.02            Prob >= chibar2 = 0.000

83.
84. xttobit ypen3 i.gender age i.gali i.educ income if country==35, ll(0) iterate(25) ba
> selevels tobit

Fitting comparison model:

Fitting constant-only model:

Iteration 0:  Log likelihood = -13059.972
Iteration 1:  Log likelihood = -8969.4753
Iteration 2:  Log likelihood = -8848.5585
Iteration 3:  Log likelihood = -8845.8105
Iteration 4:  Log likelihood = -8845.8086
Iteration 5:  Log likelihood = -8845.8086

Fitting full model:

Iteration 0:  Log likelihood = -12748.166
Iteration 1:  Log likelihood = -8956.6397
Iteration 2:  Log likelihood = -8402.0038
Iteration 3:  Log likelihood = -8347.2466
Iteration 4:  Log likelihood = -8346.5547
Iteration 5:  Log likelihood = -8346.5545

Obtaining starting values for full model:

Iteration 0:  Log likelihood = -45249.222
Iteration 1:  Log likelihood = -45212.213
Iteration 2:  Log likelihood = -45211.856
Iteration 3:  Log likelihood = -45211.856

Fitting full model:
```

```
Iteration 0:  Log likelihood = -11974.249
Iteration 1:  Log likelihood = -9486.0836  (not concave)
Iteration 2:  Log likelihood =  -8892.275
Iteration 3:  Log likelihood = -8211.8878
Iteration 4:  Log likelihood = -8201.1002
Iteration 5:  Log likelihood = -8198.0475
Iteration 6:  Log likelihood = -8198.0437
Iteration 7:  Log likelihood = -8198.0437

Random-effects tobit regression                 Number of obs     =  5,526
                                                      Uncensored =    767
Limits: Lower =    0                               Left-censored =  4,759
        Upper = +inf                              Right-censored =      0

Group variable: id                              Number of groups  =  2,837
Random effects u_i ~ Gaussian                   Obs per group:
                                                              min =      1
                                                              avg =    1.9
                                                              max =      2

Integration method: mvaghermite                 Integration pts.  =     12

                                                Wald chi2(6)      = 429.43
Log likelihood = -8198.0437                     Prob > chi2       = 0.0000

------------------------------------------------------------------------------
                                                                  ypen3 | Co
> efficient
>           Std. err.
>                          z
>                                 P>|z|
>                                           [95% con
>                                                  f. interval]
------------------------------------------------------------------------------
                                                                 gender |
                                                                   Male |
>        0
>           (base)
                                                                 Female |
> 67.89776
>            173.9935
>                         0.39
>                                 0.696
>                                        -273.1233
>                                                  408.9188
                                                                    age |  -
> 141.9375
>            8.822652
>                       -16.09
>                                 0.000
>                                        -159.2296
>                                                 -124.6454
                                                                   gali |
                                                            Not limited |
>        0
>           (base)
                                                                Limited |
> 3129.336
>            212.4268
>                        14.73
>                                 0.000
>                                         2712.987
>                                                  3545.685
                                                                   educ |
    Less than primary, primary and lower secondary education (ISCED levels 0-2) |
>        0
>           (base)
```

```
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)  |
> -653.254
>           217.7769
>                       -3.00
>                              0.003
>                                      -1080.089
>                                                 -226.419
                                              Tertiary education (ISCED levels 5-8)  |  -
> 1353.043
>           265.9793
>                       -5.09
>                              0.000
>                                      -1874.353
>                                                -831.7329
                                                                                      |
                                                                               income |  -
> .0763606
>           .0097108
>                       -7.86
>                              0.000
>                                      -.0953934
>                                                -.0573278
                                                                                _cons |
> 5497.181
>           664.9515
>                        8.27
>                              0.000
>                                         4193.9
>                                                 6800.462
--------------------------------------------------------------------------------------+---
  ------------------------------------------------------------
                                                                             /sigma_u |
>  2512.79
>           113.4388
>                       22.15
>                              0.000
>                                       2290.454
>                                                 2735.126
                                                                             /sigma_e |
> 1997.686
>           73.26852
>                       27.27
>                              0.000
>                                       1854.083
>                                                  2141.29
--------------------------------------------------------------------------------------+---
  ------------------------------------------------------------
                                                                                  rho |
> .6127313
>           .0271584
>                                       .5586039
>                                                 .6647491
--------------------------------------------------------------------------------------+---
  ------------------------------------------------------------
LR test of sigma_u=0: chibar2(01) = 297.02             Prob >= chibar2 = 0.000


85. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(CoreEE) replace
  results.xls
  dir : seeout
```

```
86. outreg2 using resultsNoStar.xls, excel dec(3) ctitle(CoreEE) noaster replace
  resultsNoStar.xls
  dir : seeout

87.
88. *for disability
89.
90. generate ageD=0

91. replace ageD=1 if age>=55 & age<=64
  (2,792 real changes made)

92. replace ageD=2 if age>=65 & age<=74
  (3,130 real changes made)

93. replace ageD=3 if age>=75 & age<=84
  (2,531 real changes made)

94. replace ageD=4 if age>=85
  (848 real changes made)

95. label define ageDl 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Bet
  > ween 65 and 74 years old" 3 "Between 75 and 84 years old" 4 "Older than 85 years"

96. label values ageD ageDl

97.
98. xttobit ypen3 i.gender i.ageD i.gali i.educ income if country==35, ll(0) iterate(25)
  >  baselevels

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -13059.972
  Iteration 1:  Log likelihood = -8969.4753
  Iteration 2:  Log likelihood = -8848.5585
  Iteration 3:  Log likelihood = -8845.8105
  Iteration 4:  Log likelihood = -8845.8086
  Iteration 5:  Log likelihood = -8845.8086

  Fitting full model:

  Iteration 0:  Log likelihood = -12648.913
  Iteration 1:  Log likelihood = -8761.5091
  Iteration 2:  Log likelihood = -8249.2098
  Iteration 3:  Log likelihood = -8214.6742
  Iteration 4:  Log likelihood = -8214.5255
  Iteration 5:  Log likelihood = -8214.5255

  Obtaining starting values for full model:

  Iteration 0:  Log likelihood =  -45078.51
  Iteration 1:  Log likelihood = -45041.929
  Iteration 2:  Log likelihood = -45041.573
  Iteration 3:  Log likelihood = -45041.573

  Fitting full model:

  Iteration 0:  Log likelihood = -11890.639
  Iteration 1:  Log likelihood =  -9420.102  (not concave)
  Iteration 2:  Log likelihood = -8432.7335
  Iteration 3:  Log likelihood = -8207.1673
  Iteration 4:  Log likelihood = -8090.1718
  Iteration 5:  Log likelihood =  -8088.846
  Iteration 6:  Log likelihood = -8088.8458

  Random-effects tobit regression                   Number of obs     =  5,526
                                                           Uncensored =    767
  Limits: Lower =    0                               Left-censored =  4,759
          Upper = +inf                              Right-censored =      0
```

```
Group variable: id                                  Number of groups  =  2,837
Random effects u_i ~ Gaussian                       Obs per group:
                                                                 min =      1
                                                                 avg =    1.9
                                                                 max =      2

Integration method: mvaghermite                     Integration pts.  =     12

                                                    Wald chi2(9)      = 601.85
Log likelihood = -8088.8458                         Prob > chi2       = 0.0000

────────────────────────────────────────────────────────────────────────────┬───
  ────────────────────────────────────────────────────────────
                                                                       ypen3 │ Co
> efficient
>           Std. err.
>                           z
>                                P>|z|
>                                          [95% con
>                                                 f. interval]
────────────────────────────────────────────────────────────────────────────┼───
  ────────────────────────────────────────────────────────────
                                                                      gender │
                                                                        Male │
>         0
>          (base)
                                                                      Female │
> 74.80538
>           159.7569
>                        0.47
>                                0.640
>                                        -238.3124
>                                                   387.9231
                                                                             │
                                                                        ageD │
                                                      Less than 55 years old │
>         0
>          (base)
                                                 Between 55 and 64 years old │
> 596.8779
>           251.9293
>                        2.37
>                                0.018
>                                         103.1057
>                                                    1090.65
                                                 Between 65 and 74 years old │  -
> 3029.058
>           279.6792
>                      -10.83
>                                0.000
>                                        -3577.219
>                                                  -2480.896
                                                 Between 75 and 84 years old │  -
> 3289.274
>           288.1068
>                      -11.42
>                                0.000
>                                        -3853.953
>                                                  -2724.595
                                                         Older than 85 years │  -
> 3271.839
>            344.263
>                       -9.50
>                                0.000
>                                        -3946.582
>                                                  -2597.096
                                                                             │
                                                                        gali │
                                                                 Not limited │
>         0
>          (base)
                                                                     Limited │
```

```
>  3124.31
>           201.2362
>                      15.53
>                             0.000
>                                     2729.894
>                                                3518.726
                                                                      |
                                                                 educ |
    Less than primary, primary and lower secondary education (ISCED levels 0-2) |
>      0
>          (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) | -
> 581.0797
>           199.3518
>                      -2.91
>                             0.004
>                                    -971.8021
>                                               -190.3574
                                          Tertiary education (ISCED levels 5-8) | -
> 1434.437
>           245.4567
>                      -5.84
>                             0.000
>                                    -1915.523
>                                               -953.3504
                                                                      |
                                                               income | -
> .0629668
>           .0087664
>                      -7.18
>                             0.000
>                                    -.0801486
>                                                -.045785
                                                                _cons | -
> 2380.608
>           364.0823
>                      -6.54
>                             0.000
>                                    -3094.197
>                                                -1667.02
----------------------------------------------------------------------+---
  ---------------------------------------------------------
                                                             /sigma_u |
> 2232.263
>           103.8189
>                      21.50
>                             0.000
>                                     2028.782
>                                                2435.744
                                                             /sigma_e |
>   1869.6
>           68.92764
>                      27.12
>                             0.000
>                                     1734.504
>                                                2004.696
----------------------------------------------------------------------+---
  ---------------------------------------------------------
                                                                  rho |
> .5877284
>           .0290497
>                                     .5300662
>                                                .6435539
----------------------------------------------------------------------+---
  ---------------------------------------------------------
LR test of sigma_u=0: chibar2(01) = 251.36            Prob >= chibar2 = 0.000
```

```
99. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(DisabilityEE) append
  results.xls
  dir : seeout

100 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityEE) noaster append
  resultsNoStar.xls
  dir : seeout

101
102 *for education
103
104 generate ageE=0

105 replace ageE=1 if age>=55 & age<=64
  (2,792 real changes made)

106 replace ageE=2 if age>=65
  (6,509 real changes made)

107 label define agel 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Olde
  > r than 64 years"

108 label values ageE agel

109
110 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==35, ll(0) iterate(25)
  >  baselevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -13059.972
  Iteration 1:  Log likelihood = -8969.4753
  Iteration 2:  Log likelihood = -8848.5585
  Iteration 3:  Log likelihood = -8845.8105
  Iteration 4:  Log likelihood = -8845.8086
  Iteration 5:  Log likelihood = -8845.8086

  Fitting full model:

  Iteration 0:  Log likelihood = -12655.689
  Iteration 1:  Log likelihood = -8767.8048
  Iteration 2:  Log likelihood = -8251.3306
  Iteration 3:  Log likelihood = -8216.2557
  Iteration 4:  Log likelihood = -8216.1028
  Iteration 5:  Log likelihood = -8216.1028

  Obtaining starting values for full model:

  Iteration 0:  Log likelihood = -45081.832
  Iteration 1:  Log likelihood = -45047.601
  Iteration 2:  Log likelihood = -45047.303
  Iteration 3:  Log likelihood = -45047.303

  Fitting full model:

  Iteration 0:  Log likelihood =   -11888.4
  Iteration 1:  Log likelihood = -9407.8171  (not concave)
  Iteration 2:  Log likelihood = -8426.0436
  Iteration 3:  Log likelihood = -8190.8384
  Iteration 4:  Log likelihood = -8090.4197
  Iteration 5:  Log likelihood = -8089.7072
  Iteration 6:  Log likelihood =  -8089.706
  Iteration 7:  Log likelihood =  -8089.706

  Random-effects tobit regression                     Number of obs     =  5,526
                                                             Uncensored =    767
  Limits: Lower =     0                                   Left-censored =  4,759
          Upper = +inf                                   Right-censored =      0
```

```
Group variable: id                                   Number of groups  =  2,837
Random effects u_i ~ Gaussian                        Obs per group:
                                                                  min =      1
                                                                  avg =    1.9
                                                                  max =      2

Integration method: mvaghermite                      Integration pts.  =     12

                                                     Wald chi2(7)      = 597.41
Log likelihood = -8089.706                           Prob > chi2       = 0.0000

---------------------------------------------------------------------------------+----
  -------------------------------------------------------------
                                                                           ypen3 | Co
> efficient
>           Std. err.
>                           z
>                                P>|z|
>                                          [95% con
>                                                  f. interval]
---------------------------------------------------------------------------------+----
  -------------------------------------------------------------
                                                                          gender |
                                                                           Male  |
>        0
>           (base)
                                                                         Female  |
> 62.35313
>             159.539
>                        0.39
>                                0.696
>                                        -250.3376
>                                                     375.0438
                                                                                 |
                                                                            ageE |
                                                         Less than 55 years old  |
>        0
>           (base)
                                                    Between 55 and 64 years old  |
> 592.9421
>            252.0415
>                        2.35
>                                0.019
>                                         98.94985
>                                                     1086.934
                                                               Older than 64 years  |  -
> 3159.591
>            262.9301
>                      -12.02
>                                0.000
>                                        -3674.924
>                                                    -2644.257
                                                                                 |
                                                                            gali |
                                                                    Not limited  |
>        0
>           (base)
                                                                        Limited  |
> 3114.176
>            201.3249
>                       15.47
>                                0.000
>                                         2719.587
>                                                     3508.766
                                                                                 |
                                                                            educ |
    Less than primary, primary and lower secondary education (ISCED levels 0-2)  |
>        0
>           (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)  |  -
> 544.0899
>            195.6608
```

```
>                       -2.78
>                               0.005
>                                        -927.5781
>                                                    -160.6016
                                          Tertiary education (ISCED levels 5-8)  |  -
> 1414.382
>          244.2873
>                       -5.79
>                               0.000
>                                        -1893.177
>                                                    -935.5882
                                                                        income  |  -
> .0623023
>          .0087611
>                       -7.11
>                               0.000
>                                        -.0794737
>                                                    -.0451309
                                                                         _cons  |  -
> 2409.678
>          363.5704
>                       -6.63
>                               0.000
>                                        -3122.263
>                                                    -1697.093
-----------------------------------------------------------------------------------+---
  ------------------------------------------------------------
                                                                      /sigma_u  |
> 2238.428
>          103.8497
>                       21.55
>                               0.000
>                                         2034.887
>                                                     2441.97
                                                                      /sigma_e  |
> 1868.205
>           68.9285
>                       27.10
>                               0.000
>                                         1733.108
>                                                     2003.303
-----------------------------------------------------------------------------------+---
  ------------------------------------------------------------
                                                                           rho  |
> .5894256
>          .0289619
>                                         .5319203
>                                                     .6450675
-----------------------------------------------------------------------------------+---
  ------------------------------------------------------------
LR test of sigma_u=0: chibar2(01) = 252.79            Prob >= chibar2 = 0.000
```

```
111 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(EducationEE) append
  results.xls
  dir : seeout

112 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(EducationEE) noaster append
  resultsNoStar.xls
  dir : seeout
```

```
113
114 *for income
115
116 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==35, ll(0) iterate(25)
  >  baselevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -13059.972
  Iteration 1:  Log likelihood = -8969.4753
  Iteration 2:  Log likelihood = -8848.5585
  Iteration 3:  Log likelihood = -8845.8105
  Iteration 4:  Log likelihood = -8845.8086
  Iteration 5:  Log likelihood = -8845.8086

  Fitting full model:

  Iteration 0:  Log likelihood = -12655.689
  Iteration 1:  Log likelihood = -8767.8048
  Iteration 2:  Log likelihood = -8251.3306
  Iteration 3:  Log likelihood = -8216.2557
  Iteration 4:  Log likelihood = -8216.1028
  Iteration 5:  Log likelihood = -8216.1028

  Obtaining starting values for full model:

  Iteration 0:  Log likelihood = -45081.832
  Iteration 1:  Log likelihood = -45047.601
  Iteration 2:  Log likelihood = -45047.303
  Iteration 3:  Log likelihood = -45047.303

  Fitting full model:

  Iteration 0:  Log likelihood =   -11888.4
  Iteration 1:  Log likelihood = -9407.8171  (not concave)
  Iteration 2:  Log likelihood = -8426.0436
  Iteration 3:  Log likelihood = -8190.8384
  Iteration 4:  Log likelihood = -8090.4197
  Iteration 5:  Log likelihood = -8089.7072
  Iteration 6:  Log likelihood =  -8089.706
  Iteration 7:  Log likelihood =  -8089.706

  Random-effects tobit regression                    Number of obs     =  5,526
                                                            Uncensored =    767
  Limits: Lower =     0                                  Left-censored =  4,759
          Upper = +inf                                  Right-censored =      0

  Group variable: id                                 Number of groups  =  2,837
  Random effects u_i ~ Gaussian                      Obs per group:
                                                                   min =      1
                                                                   avg =    1.9
                                                                   max =      2

  Integration method: mvaghermite                    Integration pts.  =     12

                                                     Wald chi2(7)      = 597.41
  Log likelihood = -8089.706                         Prob > chi2       = 0.0000

  ────────────────────────────────────────────────────────────────────────────────────┬───
    ──────────────────────────────────────────────────────────────
                                                                                ypen3 │ Co
  > efficient
  >           Std. err.
  >                               z
  >                                    P>|z|
  >                                              [95% con
  >                                                      f. interval]
  ────────────────────────────────────────────────────────────────────────────────────┼───
    ──────────────────────────────────────────────────────────────
                                                                               gender │
```

```
                                                              Male  |
>        0
>           (base)
                                                            Female  |
> 62.35313
>             159.539
>                        0.39
>                              0.696
>                                     -250.3376
>                                               375.0438
                                                                    |
                                                              ageE  |
                                              Less than 55 years old |
>        0
>           (base)
                                         Between 55 and 64 years old |
> 592.9421
>            252.0415
>                        2.35
>                              0.019
>                                      98.94985
>                                               1086.934
                                                Older than 64 years  |  -
> 3159.591
>            262.9301
>                      -12.02
>                              0.000
>                                     -3674.924
>                                              -2644.257
                                                                    |
                                                              gali  |
                                                       Not limited  |
>        0
>           (base)
                                                           Limited  |
> 3114.176
>            201.3249
>                       15.47
>                              0.000
>                                      2719.587
>                                               3508.766
                                                                    |
                                                              educ  |
   Less than primary, primary and lower secondary education (ISCED levels 0-2) |
>        0
>           (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |  -
> 544.0899
>            195.6608
>                       -2.78
>                              0.005
>                                     -927.5781
>                                              -160.6016
                                     Tertiary education (ISCED levels 5-8) |  -
> 1414.382
>            244.2873
>                       -5.79
>                              0.000
>                                     -1893.177
>                                              -935.5882
                                                                    |
                                                            income  |  -
> .0623023
>            .0087611
>                       -7.11
>                              0.000
>                                     -.0794737
>                                              -.0451309
                                                             _cons  |  -
> 2409.678
>            363.5704
>                       -6.63
```

```
>                              0.000
>                                       -3122.263
>                                                   -1697.093
------------------------------------------------------------------------------------+---
  ---------------------------------------------------------------
                                                                        /sigma_u |
> 2238.428
>          103.8497
>                     21.55
>                              0.000
>                                        2034.887
>                                                    2441.97
                                                                        /sigma_e |
> 1868.205
>           68.9285
>                     27.10
>                              0.000
>                                        1733.108
>                                                   2003.303
------------------------------------------------------------------------------------+---
  ---------------------------------------------------------------
                                                                            rho |
> .5894256
>          .0289619
>                                        .5319203
>                                                   .6450675
------------------------------------------------------------------------------------+---
  ---------------------------------------------------------------
LR test of sigma_u=0: chibar2(01) = 252.79            Prob >= chibar2 = 0.000

117 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(IncomeEE) append
  results.xls
  dir : seeout

118 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeEE) noaster append
  resultsNoStar.xls
  dir : seeout

119
120 *-------
121 *LITHUANIA
122
123 *general model
124
125 xttobit ypen3 i.gender age i.gali i.educ income if country==48, ll(0) iterate(25) ba
  > selevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -4862.8488
  Iteration 1:  Log likelihood = -2762.1949
  Iteration 2:  Log likelihood = -2652.4901
  Iteration 3:  Log likelihood = -2648.5807
  Iteration 4:  Log likelihood = -2648.5779
  Iteration 5:  Log likelihood = -2648.5779

  Fitting full model:

  Iteration 0:  Log likelihood = -4794.3696
  Iteration 1:  Log likelihood = -2581.4616
  Iteration 2:  Log likelihood = -2514.2583
  Iteration 3:  Log likelihood = -2511.8334
  Iteration 4:  Log likelihood = -2511.8237
  Iteration 5:  Log likelihood = -2511.8237

  Obtaining starting values for full model:
```

```
Iteration 0:  Log likelihood = -19482.944
Iteration 1:  Log likelihood = -19475.104
Iteration 2:  Log likelihood = -19475.077
Iteration 3:  Log likelihood = -19475.077

Fitting full model:

Iteration 0:  Log likelihood = -4426.2983
Iteration 1:  Log likelihood = -3129.6521  (not concave)
Iteration 2:  Log likelihood = -2595.7117
Iteration 3:  Log likelihood = -2441.8096
Iteration 4:  Log likelihood = -2440.2936
Iteration 5:  Log likelihood = -2439.7155
Iteration 6:  Log likelihood = -2439.7155  (backed up)
Iteration 7:  Log likelihood = -2439.7147
Iteration 8:  Log likelihood = -2439.7147

Random-effects tobit regression                  Number of obs     =  2,423
                                                        Uncensored =    218
Limits: Lower =    0                                 Left-censored =  2,205
        Upper = +inf                                Right-censored =      0

Group variable: id                               Number of groups  =  1,235
Random effects u_i ~ Gaussian                    Obs per group:
                                                               min =      1
                                                               avg =    2.0
                                                               max =      2

Integration method: mvaghermite                  Integration pts.  =     12

                                                 Wald chi2(6)      = 100.68
Log likelihood = -2439.7147                      Prob > chi2       = 0.0000

---------------------------------------------------------------------------------
  -------------------------------------------------------------
                                                                    ypen3 | Co
> efficient
>          Std. err.
>                         z
>                              P>|z|
>                                        [95% con
>                                                f. interval]
---------------------------------------------------------------------------------
  -------------------------------------------------------------
                                                                   gender |
                                                                     Male |
>       0
>          (base)
                                                                   Female |  -
> 1256.459
>           418.3478
>                      -3.00
>                              0.003
>                                      -2076.406
>                                                 -436.5129
                                                                      age |  -
> 200.0298
>           27.40675
>                      -7.30
>                              0.000
>                                       -253.746
>                                                 -146.3135
                                                                     gali |
                                                              Not limited |
>       0
>          (base)
                                                                  Limited |
>  3058.35
>            422.143
>                       7.24
```

```
>                                0.000
>                                          2230.965
>                                                     3885.735
                                                                           |
                                                                      educ |
     Less than primary, primary and lower secondary education (ISCED levels 0-2) |
>         0
>            (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> 352.9785
>           696.1143
>                        0.51
>                                0.612
>                                         -1011.381
>                                                     1717.337
                                          Tertiary education (ISCED levels 5-8) |  -
> 174.7383
>           728.8055
>                       -0.24
>                                0.811
>                                         -1603.171
>                                                     1253.694
                                                                           |
                                                                    income |  -
> .1163755
>           .0280595
>                       -4.15
>                                0.000
>                                         -.1713711
>                                                    -.0613799
                                                                     _cons |
> 6837.853
>           1965.434
>                        3.48
>                                0.001
>                                          2985.674
>                                                     10690.03
---------------------------------------------------------------------------+----
  -------------------------------------------------------------
                                                                  /sigma_u |
> 3867.504
>           311.0775
>                       12.43
>                                0.000
>                                          3257.803
>                                                     4477.204
                                                                  /sigma_e |
> 2229.672
>            165.724
>                       13.45
>                                0.000
>                                          1904.859
>                                                     2554.486
---------------------------------------------------------------------------+----
  -------------------------------------------------------------
                                                                       rho |
> .7505428
>            .037764
>                                          .6711213
>                                                     .8184268
---------------------------------------------------------------------------+----
  -------------------------------------------------------------
LR test of sigma_u=0: chibar2(01) = 144.22          Prob >= chibar2 = 0.000
```

```
126 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(CoreLT) append
  results.xls
  dir : seeout

127 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(CoreLT) noaster append
  resultsNoStar.xls
  dir : seeout

128
129 *for disability
130
131 xttobit ypen3 i.gender i.ageD i.gali i.educ income if country==48, ll(0) iterate(25)
  >  baselevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -4862.8488
  Iteration 1:  Log likelihood = -2762.1949
  Iteration 2:  Log likelihood = -2652.4901
  Iteration 3:  Log likelihood = -2648.5807
  Iteration 4:  Log likelihood = -2648.5779
  Iteration 5:  Log likelihood = -2648.5779

  Fitting full model:

  Iteration 0:  Log likelihood = -4773.3012
  Iteration 1:  Log likelihood = -2565.0925
  Iteration 2:  Log likelihood = -2484.8288
  Iteration 3:  Log likelihood = -2481.7728
  Iteration 4:  Log likelihood = -2481.7631
  Iteration 5:  Log likelihood = -2481.7631

  Obtaining starting values for full model:

  Iteration 0:  Log likelihood = -19463.113
  Iteration 1:  Log likelihood = -19454.667
  Iteration 2:  Log likelihood = -19454.636

  Fitting full model:

  Iteration 0:  Log likelihood = -4420.6504
  Iteration 1:  Log likelihood = -2956.5443  (not concave)
  Iteration 2:  Log likelihood = -2466.5563  (not concave)
  Iteration 3:  Log likelihood = -2451.0076
  Iteration 4:  Log likelihood = -2421.5766
  Iteration 5:  Log likelihood = -2420.0106
  Iteration 6:  Log likelihood = -2419.9666
  Iteration 7:  Log likelihood = -2419.9666  (backed up)
  Iteration 8:  Log likelihood = -2419.9661
  Iteration 9:  Log likelihood = -2419.9661

  Random-effects tobit regression                   Number of obs     =  2,423
                                                           Uncensored =    218
  Limits: Lower =    0                                  Left-censored =  2,205
          Upper = +inf                                 Right-censored =      0

  Group variable: id                                Number of groups  =  1,235
  Random effects u_i ~ Gaussian                     Obs per group:
                                                                  min =      1
                                                                  avg =    2.0
                                                                  max =      2

  Integration method: mvaghermite                   Integration pts.  =     12

                                                    Wald chi2(9)      = 122.93
  Log likelihood = -2419.9661                       Prob > chi2       = 0.0000
```

ypen3	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
gender						
Male	0	(base)				
Female	-1226.406	405.9354	-3.02	0.003	-2022.025	-430.7874
ageD						
Less than 55 years old	0	(base)				
Between 55 and 64 years old	1067.646	771.8018	1.38	0.167	-445.058	2580.35
Between 65 and 74 years old	-2545.243	873.6784	-2.91	0.004	-4257.622	-832.8652
Between 75 and 84 years old	-3972.701	981.3983	-4.05	0.000	-5896.206	-2049.195
Older than 85 years	-3319.867	1136.426	-2.92	0.003	-5547.222	-1092.513
gali						
Not limited	0	(base)				
Limited	3023.438	412.2912	7.33	0.000	2215.362	3831.513
educ						
Less than primary, primary and lower secondary education (ISCED levels 0-2)	0	(base)				

```
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)  |
> 396.2017
>            698.4553
>                         0.57
>                                0.571
>                                        -972.7456
>                                                     1765.149
                                           Tertiary education (ISCED levels 5-8)  |  -
> 140.2608
>            727.7094
>                        -0.19
>                                0.847
>                                        -1566.545
>                                                     1286.023
                                                                           income  |  -
> .1087979
>            .0271342
>                        -4.01
>                                0.000
>                                          -.16198
>                                                    -.0556158
                                                                            _cons  |  -
> 5421.922
>            1095.619
>                        -4.95
>                                0.000
>                                        -7569.295
>                                                    -3274.549
----------------------------------------------------------------------------------+----
  ------------------------------------------------------------
                                                                         /sigma_u  |
> 3647.767
>            297.3149
>                        12.27
>                                0.000
>                                          3065.04
>                                                     4230.493
                                                                         /sigma_e  |
> 2215.556
>            166.4085
>                        13.31
>                                0.000
>                                         1889.401
>                                                      2541.71
----------------------------------------------------------------------------------+----
  ------------------------------------------------------------
                                                                              rho  |
> .7305124
>            .0413152
>                                         .6440207
>                                                     .8049676
----------------------------------------------------------------------------------+----
  ------------------------------------------------------------
LR test of sigma_u=0: chibar2(01) = 123.59             Prob >= chibar2 = 0.000


132 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(DisabilityLT) append
  results.xls
  dir : seeout
```

```
133 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityLT) noaster append
  resultsNoStar.xls
  dir : seeout

134
135 *for education
136
137 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==48, ll(0) iterate(25)
  >  baselevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -4862.8488
  Iteration 1:  Log likelihood = -2762.1949
  Iteration 2:  Log likelihood = -2652.4901
  Iteration 3:  Log likelihood = -2648.5807
  Iteration 4:  Log likelihood = -2648.5779
  Iteration 5:  Log likelihood = -2648.5779

  Fitting full model:

  Iteration 0:  Log likelihood = -4775.9907
  Iteration 1:  Log likelihood = -2566.0898
  Iteration 2:  Log likelihood = -2486.4369
  Iteration 3:  Log likelihood = -2483.2935
  Iteration 4:  Log likelihood = -2483.2848
  Iteration 5:  Log likelihood = -2483.2848

  Obtaining starting values for full model:

  Iteration 0:  Log likelihood = -19465.758
  Iteration 1:  Log likelihood = -19457.753
  Iteration 2:  Log likelihood = -19457.725

  Fitting full model:

  Iteration 0:  Log likelihood = -4422.6177
  Iteration 1:  Log likelihood =  -2966.526  (not concave)
  Iteration 2:  Log likelihood = -2467.0794
  Iteration 3:  Log likelihood = -2428.6416  (backed up)
  Iteration 4:  Log likelihood = -2426.2572
  Iteration 5:  Log likelihood = -2422.3491
  Iteration 6:  Log likelihood = -2422.3393
  Iteration 7:  Log likelihood = -2422.3387
  Iteration 8:  Log likelihood = -2422.3383
  Iteration 9:  Log likelihood = -2422.3383

  Random-effects tobit regression                 Number of obs     =  2,423
                                                         Uncensored =    218
  Limits: Lower =    0                               Left-censored =  2,205
          Upper = +inf                              Right-censored =      0

  Group variable: id                              Number of groups  =  1,235
  Random effects u_i ~ Gaussian                   Obs per group:
                                                                min =      1
                                                                avg =    2.0
                                                                max =      2

  Integration method: mvaghermite                 Integration pts.  =     12

                                                  Wald chi2(7)      = 121.50
  Log likelihood = -2422.3383                     Prob > chi2       = 0.0000
```

```
ypen3 | Co
> efficient
>         Std. err.
>                        z
>                              P>|z|
>                                      [95% con
>                                             f. interval]

gender |
Male |
>       0
>         (base)
Female | -
> 1259.502
>         402.4633
>                    -3.13
>                          0.002
>                                 -2048.316
>                                        -470.6886

ageE |
Less than 55 years old |
>       0
>         (base)
Between 55 and 64 years old |
> 1063.061
>         776.7657
>                    1.37
>                          0.171
>                                 -459.3717
>                                        2585.494
Older than 64 years | -
> 3051.484
>         852.7176
>                    -3.58
>                          0.000
>                                 -4722.78
>                                        -1380.188

gali |
Not limited |
>       0
>         (base)
Limited |
>   3033.1
>         413.8208
>                    7.33
>                          0.000
>                                 2222.026
>                                        3844.174

educ |
Less than primary, primary and lower secondary education (ISCED levels 0-2) |
>       0
>         (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> 805.6156
>         639.0382
>                    1.26
>                          0.207
>                                 -446.8762
>                                        2058.107
Tertiary education (ISCED levels 5-8) |
> 232.6883
>         675.2926
>                    0.34
>                          0.730
>                                 -1090.861
>                                        1556.237
```

```
                                                                    income |  -
> .1103631
>           .0273208
>                      -4.04
>                             0.000
>                                      -.1639108
>                                                 -.0568153
                                                                     _cons |  -
> 5793.143
>           1082.132
>                      -5.35
>                             0.000
>                                      -7914.083
>                                                 -3672.203
---------------------------------------------------------------------------+----
  -------------------------------------------------------------
                                                                  /sigma_u |
> 3621.426
>           292.6449
>                      12.37
>                             0.000
>                                       3047.852
>                                                  4194.999
                                                                  /sigma_e |
> 2250.038
>           167.5251
>                      13.43
>                             0.000
>                                       1921.695
>                                                  2578.381
---------------------------------------------------------------------------+----
  -------------------------------------------------------------
                                                                       rho |
> .7214854
>           .0414316
>                                       .6351044
>                                                  .7964807
---------------------------------------------------------------------------+----
  -------------------------------------------------------------
LR test of sigma_u=0: chibar2(01) = 121.89            Prob >= chibar2 = 0.000


138 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(EducationLT) append
  results.xls
  dir : seeout

139 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(EducationLT) noaster append
  resultsNoStar.xls
  dir : seeout

140
141 *for income
142
143 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==48, ll(0) iterate(25)
  >  baselevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -4862.8488
  Iteration 1:  Log likelihood = -2762.1949
  Iteration 2:  Log likelihood = -2652.4901
  Iteration 3:  Log likelihood = -2648.5807
  Iteration 4:  Log likelihood = -2648.5779
  Iteration 5:  Log likelihood = -2648.5779

  Fitting full model:
```

```
Iteration 0:  Log likelihood = -4775.9907
Iteration 1:  Log likelihood = -2566.0898
Iteration 2:  Log likelihood = -2486.4369
Iteration 3:  Log likelihood = -2483.2935
Iteration 4:  Log likelihood = -2483.2848
Iteration 5:  Log likelihood = -2483.2848

Obtaining starting values for full model:

Iteration 0:  Log likelihood = -19465.758
Iteration 1:  Log likelihood = -19457.753
Iteration 2:  Log likelihood = -19457.725

Fitting full model:

Iteration 0:  Log likelihood = -4422.6177
Iteration 1:  Log likelihood =  -2966.526  (not concave)
Iteration 2:  Log likelihood = -2467.0794
Iteration 3:  Log likelihood = -2428.6416  (backed up)
Iteration 4:  Log likelihood = -2426.2572
Iteration 5:  Log likelihood = -2422.3491
Iteration 6:  Log likelihood = -2422.3393
Iteration 7:  Log likelihood = -2422.3387
Iteration 8:  Log likelihood = -2422.3383
Iteration 9:  Log likelihood = -2422.3383

Random-effects tobit regression                    Number of obs     =  2,423
                                                        Uncensored =    218
Limits: Lower =    0                                 Left-censored =  2,205
        Upper = +inf                                Right-censored =      0

Group variable: id                                 Number of groups  =  1,235
Random effects u_i ~ Gaussian                      Obs per group:
                                                                 min =      1
                                                                 avg =    2.0
                                                                 max =      2

Integration method: mvaghermite                    Integration pts.  =     12

                                                   Wald chi2(7)      = 121.50
Log likelihood = -2422.3383                        Prob > chi2       = 0.0000

---------------------------------------------------------------------------------------
                                                                            ypen3 | Co
> efficient
>          Std. err.
>                         z
>                                P>|z|
>                                          [95% con
>                                                 f. interval]
---------------------------------------------------------------------------------------
                                                                           gender |
                                                                             Male |
>        0
>           (base)
                                                                           Female |  -
> 1259.502
>            402.4633
>                      -3.13
>                                0.002
>                                        -2048.316
>                                                  -470.6886
                                                                                  |
                                                                             ageE |
                                                           Less than 55 years old |
>        0
>           (base)
                                                      Between 55 and 64 years old |
> 1063.061
>            776.7657
```

```
>                         1.37
>                                0.171
>                                         -459.3717
>                                                      2585.494
                                                             Older than 64 years  |  -
> 3051.484
>           852.7176
>                        -3.58
>                                0.000
>                                          -4722.78
>                                                     -1380.188
                                                                                  |
                                                                            gali  |
                                                                    Not limited   |
>       0
>             (base)
                                                                        Limited   |
>   3033.1
>           413.8208
>                         7.33
>                                0.000
>                                          2222.026
>                                                      3844.174
                                                                                  |
                                                                            educ  |
    Less than primary, primary and lower secondary education (ISCED levels 0-2)   |
>       0
>             (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)  |
> 805.6156
>           639.0382
>                         1.26
>                                0.207
>                                         -446.8762
>                                                      2058.107
                                          Tertiary education (ISCED levels 5-8)   |
> 232.6883
>           675.2926
>                         0.34
>                                0.730
>                                         -1090.861
>                                                      1556.237
                                                                                  |
                                                                          income  |  -
> .1103631
>           .0273208
>                        -4.04
>                                0.000
>                                         -.1639108
>                                                     -.0568153
                                                                           _cons  |  -
> 5793.143
>           1082.132
>                        -5.35
>                                0.000
>                                         -7914.083
>                                                     -3672.203
----------------------------------------------------------------------------------+----
  ------------------------------------------------------------
                                                                        /sigma_u  |
> 3621.426
>           292.6449
>                        12.37
>                                0.000
>                                          3047.852
>                                                      4194.999
                                                                        /sigma_e  |
> 2250.038
>           167.5251
>                        13.43
>                                0.000
>                                          1921.695
```

```
>                                                   2578.381
──────────────────────────────────────────────────────────────────────────┼───
  ────────────────────────────────────────────────────────────
                                                                      rho |
> .7214854
>          .0414316
>                                         .6351044
>                                                   .7964807
──────────────────────────────────────────────────────────────────────────┼───
  ────────────────────────────────────────────────────────────
LR test of sigma_u=0: chibar2(01) = 121.89            Prob >= chibar2 = 0.000


144 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(IncomeLT) append
  results.xls
  dir : seeout

145 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeLT) noaster append
  resultsNoStar.xls
  dir : seeout

146
147 *-------
148 *LATVIA
149
150 *general model
151
152 xttobit ypen3 i.gender age i.gali i.educ income if country==57, ll(0) iterate(25) ba
  > selevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -3579.9263
  Iteration 1:  Log likelihood = -1698.9624
  Iteration 2:  Log likelihood = -1616.2797
  Iteration 3:  Log likelihood = -1616.1126
  Iteration 4:  Log likelihood = -1616.1126

  Fitting full model:

  Iteration 0:  Log likelihood = -3514.5793
  Iteration 1:  Log likelihood = -1558.0582
  Iteration 2:  Log likelihood = -1466.5931
  Iteration 3:  Log likelihood = -1456.8731
  Iteration 4:  Log likelihood = -1456.7351
  Iteration 5:  Log likelihood =  -1456.735

  Obtaining starting values for full model:

  Iteration 0:  Log likelihood = -15455.982
  Iteration 1:  Log likelihood = -15453.224
  Iteration 2:  Log likelihood = -15453.219

  Fitting full model:

  Iteration 0:  Log likelihood = -3298.0239
  Iteration 1:  Log likelihood =  -1549.235
  Iteration 2:  Log likelihood =  -1517.082  (backed up)
  Iteration 3:  Log likelihood = -1445.3787
  Iteration 4:  Log likelihood = -1433.5806
  Iteration 5:  Log likelihood = -1431.6704
  Iteration 6:  Log likelihood =  -1431.668
  Iteration 7:  Log likelihood =  -1431.668

  Random-effects tobit regression                  Number of obs     =  2,003
                                                          Uncensored =    132
  Limits: Lower =    0                               Left-censored =  1,871
          Upper = +inf                              Right-censored =      0
```

```
Group variable: id                                  Number of groups  =  1,014
Random effects u_i ~ Gaussian                       Obs per group:
                                                                 min =      1
                                                                 avg =    2.0
                                                                 max =      2

Integration method: mvaghermite                     Integration pts.  =     12

                                                    Wald chi2(6)      =  94.04
Log likelihood = -1431.668                          Prob > chi2       = 0.0000

----------------------------------------------------------------------------------
  ------------------------------------------------------------
                                                                      ypen3 | Co
> efficient
>          Std. err.
>                          z
>                               P>|z|
>                                         [95% con
>                                                 f. interval]
----------------------------------------------------------------------------------
  ------------------------------------------------------------
                                                                     gender |
                                                                       Male |
>        0
>           (base)
                                                                     Female |  -
> 641.0127
>           402.2035
>                      -1.59
>                               0.111
>                                       -1429.317
>                                                  147.2917
                                                                            |
                                                                        age |  -
> 217.7174
>           29.76099
>                      -7.32
>                               0.000
>                                       -276.0479
>                                                 -159.3869
                                                                            |
                                                                       gali |
                                                                Not limited |
>        0
>           (base)
                                                                    Limited |
> 4136.484
>           584.2676
>                       7.08
>                               0.000
>                                         2991.34
>                                                  5281.627
                                                                            |
                                                                       educ |
    Less than primary, primary and lower secondary education (ISCED levels 0-2) |
>        0
>           (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> 596.2011
>           616.5269
>                       0.97
>                               0.334
>                                       -612.1695
>                                                  1804.572
                                          Tertiary education (ISCED levels 5-8) |  -
> 638.7354
>           817.3516
>                      -0.78
>                               0.435
>                                       -2240.715
>                                                  963.2443
```

```
                                                                               income |
> -.219487
>           .0454556
>                       -4.83
>                              0.000
>                                      -.3085783
>                                                  -.1303956
                                                                                _cons |
> 6995.528
>           1787.025
>                        3.91
>                              0.000
>                                       3493.023
>                                                   10498.03
--------------------------------------------------------------------------------------+---
  ------------------------------------------------------------
                                                                             /sigma_u |
> 2633.705
>           288.8006
>                        9.12
>                              0.000
>                                       2067.666
>                                                   3199.744
                                                                             /sigma_e |
> 2104.673
>           191.4314
>                       10.99
>                              0.000
>                                       1729.475
>                                                   2479.872
--------------------------------------------------------------------------------------+---
  ------------------------------------------------------------
                                                                                  rho |
> .6102737
>           .0643687
>                                       .4805192
>                                                   .7287097
--------------------------------------------------------------------------------------+---
  ------------------------------------------------------------
LR test of sigma_u=0: chibar2(01) = 50.13             Prob >= chibar2 = 0.000


153 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(Core) append
  results.xls
  dir : seeout

154 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(Core) noaster append
  resultsNoStar.xls
  dir : seeout

155
156 *for disability
157
158 xttobit ypen3 i.gender i.ageD i.gali i.educ income if country==57, ll(0) iterate(25)
  >  baselevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -3579.9263
  Iteration 1:  Log likelihood = -1698.9624
  Iteration 2:  Log likelihood = -1616.2797
  Iteration 3:  Log likelihood = -1616.1126
  Iteration 4:  Log likelihood = -1616.1126

  Fitting full model:
```

```
Iteration 0:  Log likelihood =  -3506.908
Iteration 1:  Log likelihood = -1552.0409
Iteration 2:  Log likelihood = -1457.4938
Iteration 3:  Log likelihood = -1445.8596
Iteration 4:  Log likelihood = -1445.3313
Iteration 5:  Log likelihood = -1445.2677
Iteration 6:  Log likelihood = -1445.2581
Iteration 7:  Log likelihood = -1445.2563
Iteration 8:  Log likelihood = -1445.2559
Iteration 9:  Log likelihood = -1445.2559
Iteration 10: Log likelihood = -1445.2558

Obtaining starting values for full model:

Iteration 0:  Log likelihood =  -15445.47
Iteration 1:  Log likelihood = -15441.736
Iteration 2:  Log likelihood = -15441.727

Fitting full model:

Iteration 0:  Log likelihood = -3290.4204
Iteration 1:  Log likelihood = -1565.0715  (not concave)
Iteration 2:  Log likelihood = -1487.9781
Iteration 3:  Log likelihood =  -1434.153
Iteration 4:  Log likelihood =  -1421.347
Iteration 5:  Log likelihood =  -1420.921
Iteration 6:  Log likelihood =  -1420.885
Iteration 7:  Log likelihood =   -1420.88
Iteration 8:  Log likelihood =  -1420.879
Iteration 9:  Log likelihood = -1420.8788
Iteration 10: Log likelihood = -1420.8787
Iteration 11: Log likelihood = -1420.8787

Random-effects tobit regression                     Number of obs     =  2,003
                                                           Uncensored =    132
Limits: Lower =    0                                    Left-censored =  1,871
        Upper = +inf                                   Right-censored =      0

Group variable: id                                  Number of groups  =  1,014
Random effects u_i ~ Gaussian                       Obs per group:
                                                                  min =      1
                                                                  avg =    2.0
                                                                  max =      2

Integration method: mvaghermite                     Integration pts.  =     12

                                                    Wald chi2(9)      =  94.08
Log likelihood = -1420.8787                         Prob > chi2       = 0.0000

------------------------------------------------------------------------------------
  -------------------------------------------------------------
                                                                        ypen3 | Co
> efficient
>           Std. err.
>                           z
>                               P>|z|
>                                         [95% con
>                                                 f. interval]
------------------------------------------------------------------------------------
  -------------------------------------------------------------
                                                                       gender |
                                                                         Male |
>       0
>          (base)
                                                                       Female |  -
> 341.5978
>           399.6168
>                      -0.85
>                              0.393
>                                        -1124.832
>                                                   441.6368
                                                                              |
```

ageD |
Less than 55 years old |
> 0
> (base)
Between 55 and 64 years old |
> 461.6804
> 593.136
> 0.78
> 0.436
> -700.8448
> 1624.206
Between 65 and 74 years old | -
> 2342.506
> 728.4104
> -3.22
> 0.001
> -3770.164
> -914.8481
Between 75 and 84 years old | -
> 5788.652
> 1271.983
> -4.55
> 0.000
> -8281.693
> -3295.611
Older than 85 years | -
> 15114.06
> 408420.4
> -0.04
> 0.970
> -815603.4
> 785375.3
|
gali |
Not limited |
> 0
> (base)
Limited |
> 3705.149
> 543.4916
> 6.82
> 0.000
> 2639.925
> 4770.373
|
educ |
Less than primary, primary and lower secondary education (ISCED levels 0-2) |
> 0
> (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> 156.8724
> 617.0567
> 0.25
> 0.799
> -1052.536
> 1366.281
Tertiary education (ISCED levels 5-8) | -
> 885.9491
> 815.3292
> -1.09
> 0.277
> -2483.965
> 712.0668
|
income | -
> .1950166
> .0438524
> -4.45
> 0.000
> -.2809658
> -.1090674
_cons | -

```
> 5421.555
>           990.1776
>                      -5.48
>                             0.000
>                                     -7362.267
>                                                -3480.842
──────────────────────────────────────────────────────────────────────────────┼───
  ───────────────────────────────────────────────────────────
                                                                     /sigma_u │
> 2570.351
>           279.8937
>                       9.18
>                             0.000
>                                      2021.769
>                                                 3118.933
                                                                     /sigma_e │
> 2041.861
>           189.3496
>                      10.78
>                             0.000
>                                      1670.743
>                                                  2412.98
──────────────────────────────────────────────────────────────────────────────┼───
  ───────────────────────────────────────────────────────────
                                                                          rho │
>    .6131
>           .0653765
>                                      .4811277
>                                                 .7330748
──────────────────────────────────────────────────────────────────────────────┴───
  ───────────────────────────────────────────────────────────
LR test of sigma_u=0: chibar2(01) = 48.75             Prob >= chibar2 = 0.000


159 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(DisabilityLV) append
  results.xls
  dir : seeout

160 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityLV) noaster append
  resultsNoStar.xls
  dir : seeout

161
162 *for education
163
164 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==57, ll(0) iterate(25)
  >  baselevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -3579.9263
  Iteration 1:  Log likelihood = -1698.9624
  Iteration 2:  Log likelihood = -1616.2797
  Iteration 3:  Log likelihood = -1616.1126
  Iteration 4:  Log likelihood = -1616.1126

  Fitting full model:

  Iteration 0:  Log likelihood =  -3511.125
  Iteration 1:  Log likelihood = -1555.4342
  Iteration 2:  Log likelihood = -1464.3483
  Iteration 3:  Log likelihood = -1455.0028
  Iteration 4:  Log likelihood = -1454.8874
  Iteration 5:  Log likelihood = -1454.8873

  Obtaining starting values for full model:
```

```
Iteration 0:  Log likelihood = -15450.207
Iteration 1:  Log likelihood = -15446.684
Iteration 2:  Log likelihood = -15446.676
Iteration 3:  Log likelihood = -15446.676

Fitting full model:

Iteration 0:  Log likelihood = -3293.0056
Iteration 1:  Log likelihood = -1558.0001  (not concave)
Iteration 2:  Log likelihood = -1483.8455
Iteration 3:  Log likelihood = -1440.8292
Iteration 4:  Log likelihood = -1431.4216
Iteration 5:  Log likelihood = -1431.0339
Iteration 6:  Log likelihood = -1431.0316
Iteration 7:  Log likelihood = -1431.0316

Random-effects tobit regression                 Number of obs     =  2,003
                                                       Uncensored =    132
Limits: Lower =    0                                Left-censored =  1,871
        Upper = +inf                               Right-censored =      0

Group variable: id                              Number of groups  =  1,014
Random effects u_i ~ Gaussian                   Obs per group:
                                                              min =      1
                                                              avg =    2.0
                                                              max =      2

Integration method: mvaghermite                 Integration pts.  =     12

                                                Wald chi2(7)      =  98.95
Log likelihood = -1431.0316                     Prob > chi2       = 0.0000

------------------------------------------------------------------------------
                                                                 ypen3 | Co
> efficient
>           Std. err.
>                           z
>                                P>|z|
>                                          [95% con
>                                                 f. interval]
------------------------------------------------------------------------------
                                                                gender |
                                                                  Male |
>       0
>           (base)
                                                                Female |  -
> 433.5706
>           394.5748
>                        -1.10
>                                0.272
>                                        -1206.923
>                                                    339.7818
                                                                       |
                                                                  ageE |
                                                Less than 55 years old |
>       0
>           (base)
                                           Between 55 and 64 years old |
> 383.5329
>           601.6318
>                         0.64
>                                0.524
>                                        -795.6438
>                                                     1562.71
                                                  Older than 64 years  |
> -3431.92
>           739.1043
>                        -4.64
>                                0.000
>                                        -4880.537
```

```
>                                                         -1983.302
                                                                                  |
                                                                             gali |
                                                                      Not limited |
>        0
>           (base)
                                                                          Limited |
> 3710.976
>            551.8659
>                        6.72
>                               0.000
>                                         2629.338
>                                                     4792.613
                                                                                  |
                                                                             educ |
    Less than primary, primary and lower secondary education (ISCED levels 0-2)  |
>        0
>           (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> 688.1303
>             582.771
>                        1.18
>                               0.238
>                                        -454.0798
>                                                      1830.34
                                          Tertiary education (ISCED levels 5-8) |  -
> 468.8768
>            775.7078
>                       -0.60
>                               0.546
>                                        -1989.236
>                                                     1051.483
                                                                                  |
                                                                           income |  -
> .1964301
>            .0446491
>                       -4.40
>                               0.000
>                                        -.2839408
>                                                    -.1089194
                                                                            _cons |  -
> 5825.039
>            996.1594
>                       -5.85
>                               0.000
>                                        -7777.476
>                                                    -3872.603
----------------------------------------------------------------------------------+---
  ------------------------------------------------------------
                                                                         /sigma_u |
> 2540.898
>             275.857
>                        9.21
>                               0.000
>                                         2000.228
>                                                     3081.568
                                                                         /sigma_e |
> 2131.659
>            194.8313
>                       10.94
>                               0.000
>                                         1749.796
>                                                     2513.521
----------------------------------------------------------------------------------+---
  ------------------------------------------------------------
                                                                              rho |
> .5869169
>             .064998
>                                         .4571951
>                                                      .707723
----------------------------------------------------------------------------------+---
  ------------------------------------------------------------
```

```
  LR test of sigma_u=0: chibar2(01) = 47.71             Prob >= chibar2 = 0.000


165 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(EducationLV) append
  results.xls
  dir : seeout

166 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(EducationLV) noaster append
  resultsNoStar.xls
  dir : seeout

167
168 *for income
169
170 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==57, ll(0) iterate(25)
  >  baselevels tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -3579.9263
  Iteration 1:  Log likelihood = -1698.9624
  Iteration 2:  Log likelihood = -1616.2797
  Iteration 3:  Log likelihood = -1616.1126
  Iteration 4:  Log likelihood = -1616.1126

  Fitting full model:

  Iteration 0:  Log likelihood =  -3511.125
  Iteration 1:  Log likelihood = -1555.4342
  Iteration 2:  Log likelihood = -1464.3483
  Iteration 3:  Log likelihood = -1455.0028
  Iteration 4:  Log likelihood = -1454.8874
  Iteration 5:  Log likelihood = -1454.8873

  Obtaining starting values for full model:

  Iteration 0:  Log likelihood = -15450.207
  Iteration 1:  Log likelihood = -15446.684
  Iteration 2:  Log likelihood = -15446.676
  Iteration 3:  Log likelihood = -15446.676

  Fitting full model:

  Iteration 0:  Log likelihood = -3293.0056
  Iteration 1:  Log likelihood = -1558.0001  (not concave)
  Iteration 2:  Log likelihood = -1483.8455
  Iteration 3:  Log likelihood = -1440.8292
  Iteration 4:  Log likelihood = -1431.4216
  Iteration 5:  Log likelihood = -1431.0339
  Iteration 6:  Log likelihood = -1431.0316
  Iteration 7:  Log likelihood = -1431.0316

  Random-effects tobit regression                   Number of obs     =  2,003
                                                           Uncensored =    132
  Limits: Lower =    0                                  Left-censored =  1,871
          Upper = +inf                                 Right-censored =      0

  Group variable: id                                Number of groups  =  1,014
  Random effects u_i ~ Gaussian                     Obs per group:
                                                                  min =      1
                                                                  avg =    2.0
                                                                  max =      2

  Integration method: mvaghermite                   Integration pts.  =     12

                                                    Wald chi2(7)      =  98.95
  Log likelihood = -1431.0316                       Prob > chi2       = 0.0000
```

```
──────────────────────────────────────────────────────────────────────────────────────┬───
  ──────────────────────────────────────────────────────────────
                                                                                ypen3 │ Co
> efficient
>          Std. err.
>                          z
>                               P>|z|
>                                         [95% con
>                                                 f. interval]
──────────────────────────────────────────────────────────────────────────────────────┼───
  ──────────────────────────────────────────────────────────────
                                                                               gender │
                                                                                 Male │
>        0
>          (base)
                                                                               Female │  -
> 433.5706
>           394.5748
>                      -1.10
>                              0.272
>                                       -1206.923
>                                                    339.7818
                                                                                      │
                                                                                 ageE │
                                                                Less than 55 years old │
>        0
>          (base)
                                                           Between 55 and 64 years old │
> 383.5329
>           601.6318
>                       0.64
>                              0.524
>                                       -795.6438
>                                                     1562.71
                                                                    Older than 64 years │
> -3431.92
>           739.1043
>                      -4.64
>                              0.000
>                                       -4880.537
>                                                   -1983.302
                                                                                      │
                                                                                 gali │
                                                                          Not limited │
>        0
>          (base)
                                                                              Limited │
> 3710.976
>           551.8659
>                       6.72
>                              0.000
>                                        2629.338
>                                                    4792.613
                                                                                      │
                                                                                 educ │
    Less than primary, primary and lower secondary education (ISCED levels 0-2) │
>        0
>          (base)
Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) │
> 688.1303
>            582.771
>                       1.18
>                              0.238
>                                       -454.0798
>                                                     1830.34
                                      Tertiary education (ISCED levels 5-8) │  -
> 468.8768
>           775.7078
>                      -0.60
>                              0.546
>                                       -1989.236
>                                                    1051.483
```

```
                                                                   income |  -
> .1964301
>           .0446491
>                      -4.40
>                             0.000
>                                    -.2839408
>                                               -.1089194
                                                                    _cons |  -
> 5825.039
>           996.1594
>                      -5.85
>                             0.000
>                                    -7777.476
>                                               -3872.603
------------------------------------------------------------------------------+---
  ---------------------------------------------------------
                                                                 /sigma_u |
> 2540.898
>            275.857
>                       9.21
>                             0.000
>                                     2000.228
>                                                3081.568
                                                                 /sigma_e |
> 2131.659
>           194.8313
>                      10.94
>                             0.000
>                                     1749.796
>                                                2513.521
------------------------------------------------------------------------------+---
  ---------------------------------------------------------
                                                                      rho |
> .5869169
>            .064998
>                                     .4571951
>                                                 .707723
------------------------------------------------------------------------------+---
  ---------------------------------------------------------
LR test of sigma_u=0: chibar2(01) = 47.71            Prob >= chibar2 = 0.000


171 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(IncomeLV) append
  results.xls
  dir : seeout

172 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeLV) noaster append
  resultsNoStar.xls
  dir : seeout

173
174 log close SimulationBaltic
        name:  SimulationBaltic
         log:  C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
  > tput.smcl
    log type:  smcl
   closed on:  25 Aug 2026, 10:51:40
----------------------------------------------------------------------------------
```